

RENJIE ZHAO | CURRICULUM VITAE

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Department of Computer Science at Johns Hopkins University

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RESEARCH INTEREST

- **Wireless Systems and Networking:** next-generation wireless network architectures (5G millimeter-wave, 6G joint communication and sensing, Internet of Things); novel radio hardware and software design (software defined radio, wireless brain interfaces, low-power ultra-wide-band).
- **Mobile and Ubiquitous Computing:** ubiquitous communication and sensing systems (smart homes, virtual/augmented reality, localization, ultra-reliable RFID for supply chains)

EDUCATION

University of California San Diego (UCSD)

Ph.D., Electrical Engineering

Advisor: Professor Xinyu Zhang

San Diego, CA, US

Sept. 2018 - Aug. 2023

University of California San Diego (UCSD)

M.S., Electrical Engineering

San Diego, CA, US

Sept. 2018 - June 2020

Shanghai Jiao Tong University (SJTU)

B.E., Electric Power Engineering and Automation

Shanghai, China

Sept. 2014 - June 2018

EMPLOYMENT

Johns Hopkins University

Assistant Professor.

Baltimore, US

Aug. 2023 - Present

Microsoft Research

Researcher Intern. Host: Krishna Chintalapudi

Remote, US

June 2021 - Dec. 2021

Alibaba Group

Research Intern. Host: Pengyu Zhang, Yunfei Ma

Remote, US

July 2020 - Sept. 2020

Alibaba Group

Research Intern. Host: Pengyu Zhang, Yunfei Ma

Bellevue, WA, US

Sept. 2019 - Jan. 2020

UCSD

Graduate Student Researcher. Advisor: Professor Xinyu Zhang

La Jolla, CA, US

Sept. 2018 - June 2023

PUBLICATIONS

Note: '*' marks co-primary authors.

Conference Papers:

[C9] "SAPE: Demystifying Sub-band Aware Power-Equalization for Cellular Networks"

Dingyu Zhang, Ish Kumar Jain, and Renjie Zhao

The 27th International Workshop on Mobile Computing Systems and Applications (HotMobile), 2026

[C8] "3in1: Multi-tone Joint Powering, Clocking, and Communication for Passive IoT"

Ruirong Huang, and Renjie Zhao

ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys), 2026

(48 papers accepted out of 257, acceptance ratio: 18.7%)

[C7] "NanoScatter: Towards Ambient IoT"

Fengyuan Zhu, Jiaqi Shen, Wenhui Li, Jianyu Luo, Renjie Zhao, Linling Zhong, Bingbing Wang, and Xiaohua Tian

ACM International Conference on Mobile Computing and Networking (MobiCom), 2025

(41 papers accepted out of 240, acceptance ratio: 17.1%)

[C6] "Enhancing mmWave Radar Sensing Using a Phased-MIMO Architecture"

Kai Zheng, Wuqiong Zhao, Timothy Woodford, Renjie Zhao, Xinyu Zhang, and Yingbo Hua

ACM International Conference on Mobile Systems, Applications and Services (**MOBISYS**), 2024
(43 papers accepted out of 263, acceptance ratio: 16.3%)

[C5] “SlimWiFi: Ultra-Low-Power IoT Radio Architecture Enabled by Asymmetric Communication”
Renjie Zhao, Kejia Wang, Kai Zheng, Xinyu Zhang, and Leung Vincent
20th USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2023
(46 out of 288 submissions (Fall), acceptance ratio: 16.0%)

[C4] “RF-Chord: Towards Deployable RFID Localization System for Logistic Networks”
Bo Liang*, Purui Wang*, **Renjie Zhao**, Pengyu Zhang, Xinyu Zhang, Hongqiang Harry Liu and Chenren Xu
20th USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2023
(50 out of 272 submissions (Spring), acceptance ratio: 18.4%)

[C3] “NFC+: Breaking NFC Networking Limits through Resonance Engineering”
Renjie Zhao*, Purui Wang*, Yunfei Ma, Pengyu Zhang, Hongqiang Harry Liu, Xianshang Lin, Xinyu Zhang, Chenren Xu and Ming Zhang
Annual conference of the ACM Special Interest Group on Data Communication on the applications, technologies, architectures, and protocols for computer communication (SIGCOMM), 2020
(54 out of 250 submissions, acceptance ratio: 21.6%)

[C2] “M-Cube: A Millimeter-Wave Massive MIMO Software Radio”
Renjie Zhao, Timothy Woodford, Teng Wei, Kun Qian and Xinyu Zhang
ACM International Conference on Mobile Computing and Networking (MobiCom), 2020
(62 out of 384 submissions, acceptance ratio: 16.1%)
Best Paper Award (2 out of 384 submission);
Open source research platform, *M-Cube website*, used by 20+ research groups

[C1] “OFDMA-Enabled Wi-Fi Backscatter”
Renjie Zhao, Fengyuan Zhu, Siyuan Peng, Yuda Feng, Xiaohua Tian, Hui Yu and Xinbing Wang
ACM International Conference on Mobile Computing and Networking (MobiCom), 2019
(55 out of 290 submissions, acceptance ratio: 19.0%)

Journal:

[J9] “RF-HOI: Recognize Human-Object Interaction with Radio Frequency Signals”
Kai Sun, Shuai Wang, **Renjie Zhao**, Ruofeng Liu, Weiwei Chen, Zhimeng Yin, Wei Gong, Shuai Wang
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp), vol. 9, no. 1, Article 18., Mar. 2025, pp. 1-24

[J8] “NanoScatter: An IC-System Co-design Towards Ambient IoT” (Invited, Highlighted)
Fengyuan Zhu, Jiaqi Shen, Wenhui Li, Jianyu Luo, **Renjie Zhao**, Linling Zhong, Bingbing Wang, and Xiaohua Tian
GetMobile: Mobile Computing and Communications, Volume 30, Issue 1, Mar. 2026, pp 40–33

[J7] “RISensing: Leveraging Reconfigurable Intelligent Surfaces to Empower Wi-Fi Sensing”
Xiaojing Wang, Binbin Xie, Guanghui Lv, Boyang Liu, Chenhao Ma, **Renjie Zhao**, Chao Feng, and Xiaojiang Chen
IEEE Transactions on Mobile Computing, vol. 24, no. 11, Nov. 2025, pp. 12359-12374

[J6] “WULoc: Achieving Extremely Long-range High-precision Localization via Wi-Fi-UWB Connection”
Kai Sun, Shuai Wang, **Renjie Zhao**, Ruofeng Liu, Weiwei Chen, Zhimeng Yin, Wei Gong, Shuai Wang
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp), vol. 9, no. 1, Article 18., Mar. 2025, pp. 1-24

[J5] “Enabling OFDMA in Wi-Fi Backscatter”
Fengyuan Zhu, **Renjie Zhao**, Bingbing Wang, Xinbing Wang, Xinping Guan, Chenghu Zhou, and Xiaohua Tian
IEEE/ACM Transactions on Networking, vol. 32, no. 1, Feb. 2024, pp. 427-444

[J4] “M-CUBE: A Millimeter-Wave Massive MIMO Software Radio” (Invited, Highlighted)
Renjie Zhao, Timothy Woodford, Teng Wei, Kun Qian, and Xinyu Zhang
GetMobile: Mobile Computing and Communications, Volume 25, Issue 1, Mar. 2021, pp 30–33

[J3] “Synthesis of CuInS₂ nanowire arrays via solution transformation of Cu₂S self-template for enhanced photoelectrochemical performance”
Ming Li, **Renjie Zhao**, Yanjie Su, Jing Hu, Zhi Yang, and Yafei Zhang

Applied Catalysis B: Environmental, Volume 203, Apr. 2017, pp 715-724

[J2] "Hierarchically CuInS₂ Nanosheet-Constructed Nanowire Arrays for Photoelectrochemical Water Splitting"

Ming Li, **Renjie Zhao**, Yanjie Su, Jing Hu, Zhi Yang, and Yafei Zhang

Advanced Materials Interfaces, Volume 3, Issue 20, Oct. 2016, 1600494

[J1] "Carbon Quantum Dots Decorated Cu₂S Nanowire Arrays for Enhanced Photoelectrochemical Performance"

Ming Li, **Renjie Zhao**, Yanjie Su, Zhi Yang, and Yafei Zhang

Nanoscale, Volume 8, Issue 16, 2016, pp 8559-8567

Patents:

[P1] "A carbon nanotube/indium sulfide heterojunction nano wire and a preparing method thereof"

Yanjie Su, **Renjie Zhao**, Ming Li, Yafei Zhang, Zhi Yang,

Patent of invention (publicated), Publication Number : CN106391054(A), Feb. 2017

GRANTS

• **NVIDIA Academic Grant, PI**

Feb 2026

Project title: "Use Your X-ray Eye: Empower Robots for Non-line-of-sight Sensing"

• **National Institute on Aging (NIA), AI/Tech + Aging Pilot Award, PI**

May 2025 - May 2026

Project title: "In-Home Assessment of Muscle Deterioration with AI-Enhanced Smartphone Sonometry"

Co-PI: Prof. Xinyu Zhang

HONORS AND AWARDS

• ACM MobiCom'24 Best Runner-up Poster Awards

2024

• ACM MobiCom'20 Best Paper Award

2020

COURSE

- EN.601.716 Advanced Topics in Internet of Things, Fall 2026
- EN.601.416/616 Embedded Systems & Wireless IoT, Spring 2026
- EN.601.716 Advanced Topics in Internet of Things, Fall 2025
- EN.601.414/614 Computer Networks, Spring 2025
- EN.601.414/614 Computer Networks, Fall 2024
- EN.601.716 Advanced Topics in Internet of Things, Spring 2024

PRESENTATIONS

• **Highly Reliable Communication and Sensing for Battery-free IoT**

Visit talks at SJTU, THU, PKU, NJU, USTC, HUST, SEU, ZJU, MSRA, NWU, PolyU, CUHK, HKU, FDU, Nov. and Dec. 2023

• **Tutorial on M-Cube RF front-end and phased arrays**

Tutorial at the M-Cube User workshop, La Jolla, August 2023

• **SlimWiFi: Ultra-Low-Power IoT Radio Architecture Enabled by Asymmetric Communication**

Conference talk at NSDI, Boston, April 2023

• **M-Cube: A Millimeter-Wave Massive MIMO Software Radio**

Conference talk at MobiCom, Virtual, September 2020

• **NFC+: Breaking NFC Networking Limits**

Conference talk at SIGCOMM, Virtual, August 2020

• **OFDMA-Enabled Wi-Fi Backscatter**

Conference talk at SIGCOMM, Los Cabos, October 2019

PROFESSIONAL SERVICES

- TPC @ MobiCom 2025, 2026; NSDI 2025; MobiSys 2025, 2026; HotMobile 2024, 2025, 2026; ENSSys 2025; WiNTECH 2025
- MobiSys 2025 Publicity Chair
- HotMobile 2025 Student Travel Grant Co-Chair
- MobiCom 2024 MobiJob Event Chair
- SenSys Shadow PC, 2022
- Reviewer of IEEE/ACM Transactions on Networking, ACM Transactions on Internet of Things, IEEE Transactions on Wireless Communications, ACM Transactions on Sensor Networks, IEEE Internet Computing